Same-step linearization using the increment polynomial group

Project proof draft: /root/astra_v14_baseline_theta

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Status. This is a proposed local replacement for the same-step linearization construction in GTZ Appendix C. It is not an independent acceptance. It uses the bounded normalization and tautification proved in the separately frozen project unit [2], and the published algebraic polynomial-map results specified below. It does not assume GI(s).

1 The required conclusion

Let G/Γ be a connected simply connected nilpotent Lie group modulo a lattice, with a rational degree filtration $G_\bullet$ of degree at most $s \geq 1$. Every G_i is connected and rational. Let a scalar polynomial nilsequence be $F(g(n)\Gamma)$, with F complex-valued, 1-bounded and Lipschitz. The polynomial g is adapted to $G_\bullet$ in the right-difference convention $\partial_u g(n) = g(n+u)g(n)^{-1}$.

Theorem 1. *For this fixed filtered nilmanifold there is a fixed s-step nilmanifold $\widetilde{G}/\widetilde{\Gamma}$ with a fixed smooth metric, and a constant C depending only on the original fixed geometric data, with the following property. For every polynomial g and every such F, there are $a \in \widetilde{G}$ and a 1-bounded Lipschitz function $\widetilde{F} : \widetilde{G}/\widetilde{\Gamma} \longrightarrow \mathbb{C}$ such that*

$$F(g(n)\Gamma) = \widetilde{F}(a^n\widetilde{\Gamma}) \quad (n \in \mathbb{Z}), \qquad \mathrm{Lip}(\widetilde{F}) \leq C\,\mathrm{Lip}(F).$$

Fixed smooth metric comparisons are included in C. The same construction applies to limit polynomial maps and bounded Lipschitz limit functions, with fixed nilmanifold and uniform constants. Consequently a finite-family polynomial inverse theorem of degree s implies the finite-family linear inverse theorem of step s, without changing the input Gowers norm or losing correlation.

If $\mathrm{Lip}(F) = 0$, the constant test function has a constant extension and the displayed bound is understood literally. A point nilmanifold can be used for a constant sequence. Step at most s is the meaning of s-step throughout.

By [2, Theorem 1], restrict to a fixed taut representation with $G_0 = G_1 = G$ and replace the polynomial by h, where $h(0) = 1$. The exact identity is

$$g(0) = q\gamma, \qquad h(n) = q^{-1}g(n)\gamma^{-1}, \qquad F_q(x\Gamma) = F(qx\Gamma), \qquad F_q(h(n)\Gamma) = F(g(n)\Gamma).$$

Here q lies in a fixed compact set and $\mathrm{Lip}(F_q)$ is bounded by a fixed multiple of $\mathrm{Lip}(F)$. The underlying taut filtered nilmanifold is fixed independently of g. It therefore suffices to prove the theorem for this normalized h and F_q. In what follows all geometric data refer to this fixed taut representation.

2 Polynomial groups and their Lie algebras

For $i \geq 0$ put

$$P_i = \mathrm{poly}(\mathbb{Z}_{\mathbb{N}} \longrightarrow G_{\bullet+i}).$$

The zeroth group of the shifted filtration is G_i; it need not be taut. The group operation is pointwise multiplication. By [1, Corollary B.4 and Proposition B.6], these are groups with

$$[P_i, P_j] \subseteq P_{i+j}, \qquad P_i \triangleleft P_0. \tag{1}$$

Those results concern abstract filtered groups and require no equality of the first two groups. Their conditions hold because the shifted groups inherit nesting and the commutator law from $G_\bullet$.

Here is an explicit finite-dimensional description, including the smooth real shift needed below. Let $\mathfrak{g}_i = \log G_i$ and define

$$V_i = \left\{ X(t) = \sum_{j=0}^{s-i} X_j t^j : X_j \in \mathfrak{g}_{j+i} \right\} \quad (0 \leq i \leq s), \qquad V_i = 0 \quad (i > s).$$

Use pointwise Lie bracket. Bracketing coefficients gives

$$[V_i, V_j] \subseteq V_{i+j}. \tag{2}$$

For instance, a bracket of a degree-a coefficient in $\mathfrak{g}_{a+i}$ and a degree-b coefficient in $\mathfrak{g}_{b+j}$ belongs to $\mathfrak{g}_{a+b+i+j}$ and has polynomial degree $a+b$. Thus V_i is a finite-dimensional nilpotent Lie algebra for every i. Its simply connected group is obtained by the terminating BCH formula, or equivalently by pointwise exponentiation.

Lemma 2. *The pointwise exponential identifies that group with P_i. In particular P_i is connected and simply connected, and its smooth topology can be described either by the coefficients of* $\log p(t)$ *or by the Taylor coefficients in [1, Lemma B.9].*

Proof. If $i > s$, the zeroth group of the shifted filtration is trivial, so both P_i and V_i are trivial. Fix $0 \leq i \leq s$ and put $m = s - i$. We use the connectedness of every G_j stated in the setup.

First recall explicitly why the global exponential coordinates used here are smooth. For a connected nilpotent Lie group H, let $Q_a(x) = x^a$, where a is a positive integer. In left tangent trivializations, differentiating the a factors gives

$$DQ_a(x) = \sum_{k=0}^{a-1} \mathrm{Ad}(x)^{-k}.$$

The infinitesimal adjoint action lowers the lower central series. An exponential neighborhood generates H, so $\mathrm{Ad}(x)$ induces the identity on each successive quotient of that series. Consequently $DQ_a(x)$ induces multiplication by a on each quotient and is invertible. The usual local Lie exponential is smooth with derivative the identity at zero. Given $X \in \mathfrak{h}$, choose a so large that X/a lies in this local chart. The identity

$$\exp_H(Y) = Q_a(\exp_H(Y/a))$$

then proves that the exponential has invertible derivative near X. For simply connected nilpotent H, its global bijectivity is the exponential homeomorphism recalled in [1, Section 3]. Its local smooth inverses therefore agree to give a smooth global inverse. This argument uses the local Lie

exponential, differentiation of group operations and the inverse function theorem. It does not infer smoothness from the word *homeomorphism*.

By [1, Proposition B.6], P_i is a group under pointwise multiplication. By [1, Lemma B.9], its elements have the unique increasing-order Taylor expansion

$$p(n) = \prod_{j=0}^{m} u_j^{\binom{n}{j}}, \qquad u_j \in G_{i+j}, \qquad n \in \mathbb{Z}.$$

The coefficients are recovered by the triangular formulas

$$u_0 = p(0), \qquad u_j = \left(\prod_{k<j} u_k^{\binom{j}{k}} \right)^{-1} p(j).$$

The order of the displayed inverse product is part of the formula. Transfer the product manifold structure of $\prod_{j=0}^{m} G_{i+j}$ to P_i using these coordinates. Each evaluation at an integer is smooth. Evaluation of a product or inverse followed by the recovery formulas proves smoothness of the group operations. The recovered coefficients lie in the prescribed embedded subgroups; smoothness into those subgroups follows from their local embedded charts. The same formulas identify this topology with the pointwise topology.

Evaluation at $0, \dots, m$ is an injective group homomorphism $P_i \longrightarrow G^{m+1}$. Hence P_i is nilpotent, with class at most the class of G. Each closed connected subgroup of a simply connected nilpotent group is simply connected. In exponential coordinates it is the exponential of its Lie subalgebra: the finite nilpotent BCH formula makes that exponential image a connected subgroup, and uniqueness of the connected subgroup with that Lie algebra gives the assertion. Thus the product manifold above, and therefore P_i, is connected and simply connected. This step does not assert that P_0 has class at most s.

At the identity, differentiating the Taylor parametrization gives

$$(X_0, \dots, X_m) \longmapsto \left(n \longmapsto \sum_{j=0}^{m} \binom{n}{j} X_j \right), \qquad X_j \in \mathfrak{g}_{i+j}.$$

The binomial-to-monomial matrix is triangular with nonzero diagonal. A contribution to degree b comes only from $j \geq b$, and $\mathfrak{g}_{i+j} \subseteq \mathfrak{g}_{i+b}$. Descending triangular inversion preserves these same coefficient spaces. This identifies the tangent space with exactly V_i. Evaluation maps are smooth homomorphisms, so their derivatives preserve Lie brackets. The finitely many evaluations just used determine every tangent vector. They therefore identify the bracket with the pointwise bracket on V_i.

Uniqueness of one-parameter subgroups now gives

$$\mathrm{ev}_n(\exp_{P_i} X) = \exp_G(X(n)) \qquad (n \in \mathbb{Z}).$$

Apply the smooth-global-exponential argument above to P_i. Its inverse assigns to p exactly the Lie polynomial whose values are $\log_G(p(n))$. Conversely every $X \in V_i$ gives an element of P_i. This proves the pointwise exponential identification and the smooth logarithmic coefficient coordinates. The group law is determined by multiplication in G at the integer arguments. Under the ordinary nilpotent BCH identification it is the pointwise BCH law. The coordinate spaces and finite evaluation maps depend only on the fixed filtration and i. □

For real u, define T^u on V_i by $T^u X(t) = X(t+u)$. Coefficient expansion and nesting show that this preserves V_i. It is a smooth one-parameter group of Lie algebra automorphisms, since the bracket is pointwise. Exponentiating defines a smooth automorphism group on P_i, which at integer u is the ordinary shift of polynomial sequences. Its infinitesimal action is ordinary differentiation, and

$$\frac{d}{dt} V_i \subseteq V_{i+1}. \tag{3}$$

Indeed the coefficient of t^j in $X'(t)$ belongs to $\mathfrak{g}_{j+1+i}$.

3 The lattice and the same-step semidirect product

Define

$$\Lambda_i = \{p \in P_i : p(n) \in \Gamma \text{ for every } n \in \mathbb{Z}\}.$$

It is a subgroup by pointwise multiplication. We give the needed lattice argument to avoid relying on the disputed step estimate in Appendix C.

Lemma 3. *Λ_i is a discrete cocompact subgroup of P_i. Integer shifts preserve Λ_i.*

Proof. It is enough to consider $i \le s$ and put $d = s - i$; for larger i both groups are trivial. Discreteness follows from evaluation at $0, \dots, d$. A lattice-valued polynomial close enough to the identity at these finitely many points equals the identity at all those points, since Γ is discrete. Taylor uniqueness then makes it the identity polynomial.

For cocompactness, use induction to write

$$p = \gamma_j h_j c_j,$$

where $\gamma_j \in \Lambda_i$, c_j ranges over a fixed compact subset of P_i, and h_j is the identity at $0, \dots, j-1$. For $j = 0$ take $\gamma_0 = c_0 = 1$ and $h_0 = p$. Taylor expansion shows $h_j(j) \in G_{j+i}$. Choose a fixed compact left fundamental set for the lattice $G_{j+i} \cap \Gamma$ in G_{j+i} and write $h_j(j) = \lambda_j v_j$, where λ_j is in that lattice and v_j is in the compact set. Define

$$\ell_j(t) = \lambda_j^{\binom{t}{j}}, \qquad v_j^*(t) = v_j^{\binom{t}{j}}.$$

Both are in P_i, by the same one-parameter difference check as above; $\ell_j \in \Lambda_i$ because $\binom{n}{j}$ is an integer for every $n \in \mathbb{Z}$. Now put

$$\gamma_{j+1} = \gamma_j \ell_j, \qquad h_{j+1} = \ell_j^{-1} h_j (v_j^*)^{-1}, \qquad c_{j+1} = v_j^* c_j.$$

These formulas preserve the factorization and make h_{j+1} the identity at $0, \dots, j$. The possible v_j^* form a compact subset of P_i, so the possible c_{j+1} remain in a fixed compact set. At $j = d+1$, Taylor uniqueness gives $h_{d+1} = 1$. Thus $p = \gamma c$ with c in a fixed compact set, which proves left cocompactness; inversion gives right cocompactness. Finally an integer shift preserves both polynomiality and all integer lattice values, and its inverse does also. □

Now use the *increment* group P_1, not P_0, and set

$$\widetilde{G} = P_1 \rtimes_T \mathbb{R}, \qquad (p,u)(q,v) = (p\, T^u q, u+v), \qquad \widetilde{\Gamma} = \Lambda_1 \rtimes_T \mathbb{Z}. \tag{4}$$

The real action just constructed makes $\widetilde{G}$ a connected simply connected Lie group, diffeomorphic to $P_1 \times \mathbb{R}$. Its Lie algebra is $V_1 \rtimes \mathbb{R}\tau$, where $[\tau, X] = X'$. Give this Lie algebra the filtration

$$\mathfrak{l}_0 = \mathfrak{l}_1 = V_1 \oplus \mathbb{R}\tau, \qquad \mathfrak{l}_j = V_j \quad (j \ge 2).$$

Equations (2) and (3) prove $[\mathfrak{l}_i, \mathfrak{l}_j] \subseteq \mathfrak{l}_{i+j}$ for all indices. In particular $\mathfrak{l}_{s+1} = 0$. Thus $\widetilde{G}$ is nilpotent of step at most s. This explicit shifted-filtration calculation is the same-step bound used in the theorem.

The subgroup $\widetilde{\Gamma}$ is discrete, because its clock coordinate is integral and Λ_1 is discrete. It is cocompact as well. Given (p, u), left-multiply by $(1, -\lfloor u \rfloor)$, then left-reduce the resulting polynomial coordinate modulo Λ_1 using Lemma 3. The resulting representative lies in the product of a fixed compact set in P_1 and $[0, 1]$. This proves left cocompactness; inversion proves right cocompactness. The operations used for left reduction belong to $\widetilde{\Gamma}$ by the integer-shift assertion of the lemma.

4 The orbit identity and fixed compact fiber

Let $\theta : \widetilde{G}/\widetilde{\Gamma} \longrightarrow \mathbb{R}/\mathbb{Z}$ be the map induced by the clock coordinate, and set $K = \theta^{-1}(0)$. It is a fixed compact submanifold canonically identified with P_1/Λ_1: every element of the fiber can be written $(p, 0)\widetilde{\Gamma}$, and two such representatives agree precisely when their polynomial coordinates differ on the right by an element of Λ_1. Evaluation at zero gives a smooth map

$$\pi : K \longrightarrow G/\Gamma, \qquad \pi((p, 0)\widetilde{\Gamma}) = p(0)\Gamma. \tag{5}$$

It is well-defined because $\lambda(0) \in \Gamma$ for $\lambda \in \Lambda_1$. Since K is compact, π is Lipschitz for fixed smooth metrics.

The normalized polynomial $h \in P_0$ has $h(0) = 1$. Define its right increment by

$$b(t) = h(t)^{-1} h(t + 1).$$

The left increment $h(t + 1)h(t)^{-1}$ lies in P_1 directly from the iterated-difference definition. Proposition B.6 gives $P_1 \triangleleft P_0$, so conjugating that left increment by $h(t)^{-1}$ shows $b \in P_1$. This verifies the orientation of the increment actually used, rather than substituting a left increment into a right-telescoping product.

Put $a = (b, 1) \in \widetilde{G}$. For every integer n define $b_n(t) = h(t)^{-1} h(t+n)$. The exact cocycle identity

$$b_{n+m}(t) = b_n(t)\, b_m(t + n)$$

and $b_0 = 1$ prove, for positive and negative integers alike,

$$a^n = (b_n, n).$$

Modulo $\widetilde{\Gamma}$, right multiplication by $(1, -n)$ gives $a^n\widetilde{\Gamma} = (b_n, 0)\widetilde{\Gamma} \in K$. Consequently

$$\pi(a^n\widetilde{\Gamma}) = b_n(0)\Gamma = h(n)\Gamma, \qquad n \in \mathbb{Z}. \tag{6}$$

The last equality uses the normalization $h(0) = 1$.

5 Test functions, uniformity and the inverse-theorem join

The function $F_q \circ \pi$ is 1-bounded on K and has a Lipschitz constant bounded by a fixed multiple L of $\mathrm{Lip}(F)$. Extend its real and imaginary parts separately by the explicit McShane formula

$$E_u(x) = \inf_{y \in K} \big(u(y) + L\, d(x, y)\big).$$

This equals u on K and is L-Lipschitz: the triangle inequality proves the Lipschitz assertion, and the original Lipschitz inequality proves equality at points of K. Clip each extension to $[-1, 1]$,

combine them as a complex function, and apply metric projection onto the closed unit disk. This projection Q is 1-Lipschitz: the nearest-point inequalities for the disk give

$$\langle x - Qx, Qy - Qx \rangle \le 0, \qquad \langle y - Qy, Qx - Qy \rangle \le 0,$$

and therefore $\|Qx - Qy\|^2 \le \langle Qx - Qy, x - y \rangle \le \|Qx - Qy\| \|x - y\|$. It also fixes the original values on K. Thus the result $\widetilde{F}$ is 1-bounded, agrees with $F_q \circ \pi$ on K, and is at most $\sqrt{2}L$-Lipschitz. Equation (6) proves the exact equality in Theorem 1.

The nilmanifold, lattice, compact fiber, evaluation map and all metric constants depend only on the fixed original filtered nilmanifold and s. The possibly unbounded polynomial coefficients occur only in the allowed translation element a and in the bounded representative q affecting the test function. They do not alter the geometry or its Lipschitz bound. For limit data the same algebraic formulas and one fixed bounded representative section give limit choices of b and a. Apply the explicit extension formula to each representative test function. Its uniform bound and Lipschitz constant give a bounded Lipschitz limit function on the same standard nilmanifold. The ordinary orbit identities hold at every member of the defining sequences and therefore at every limit integer.

Suppose now that a polynomial inverse theorem supplies, for fixed s and δ, a finite family of rational degree-s filtered nilmanifolds, a fixed Lipschitz bound, and a positive correlation lower bound against 1-bounded test functions. Apply the theorem to each member of that finite family, taking the maximum of its finitely many fixed metric constants. There is still a finite family, now of step at most s with linear orbits, and the correlation is unchanged because the represented sequences are equal pointwise. No comparison of Gowers norms is involved in this join. This proves the final assertion of Theorem 1.

The construction also covers a point input group: P_1 is trivial and the clock group yields a circle with a constant test function, or it may be replaced by a point. For $s = 1$ the increment group consists of constant polynomials in an abelian group and the shift action is trivial, so $\widetilde{G}$ is abelian, as the required bound predicts.

6 Comparison with the failed full-group construction

The original Appendix C used $P_0 \rtimes \mathbb{R}$. For the taut abelian degree-s filtration on $\mathbb{R}$, the commutator of its clock with $p(t) = \frac{1}{2}\binom{t}{s}$, repeated s times, is the nonzero constant $1/2$. That is a commutator of length $s+1$, so that group is not s-step. This adverse example is recorded in the separate finding packet. The present construction uses P_1 and its increment b, reducing the polynomial degree available in the clock action by one. Its Lie filtration above verifies the required step bound for all inputs, not only this example. It changes the construction, while retaining the same-step linear conclusion needed by GI(s).

The finite-family polynomial inverse theorem is still a separate premise. This unit does not prove its rank reduction, symmetrization, or ultralimit-to-finite-family extraction. Those dependencies remain explicit in the theorem assembly.

References

[1] B. Green, T. Tao and T. Ziegler, *An inverse theorem for the Gowers $U^{s+1}[N]$-norm*, Annals of Mathematics **176** (2012), 1231–1372. Only the specified algebraic polynomial-map and Lie-group facts are used; the Appendix C step claim is replaced here.

[2] Project proof, *Tautification and the shifted nilcharacter construction*, draft 003, 7 November 2026. The exact author manifest and independent scope decision are pinned in the accompanying application record.